those partitions. Each node stores a unique portion of the key-space in memory. If a node fails, the node is replaced, but the data is lost. You cannot back up the data stored in Memcached. Figure 11.4 shows a Memcached cluster deployment. Remember that a VPC is a way to define a private network on AWS. A subnet is a way to separate concerns inside the VPC. The cluster nodes are distributed among multiple subnets to increase availability. The client communicates with the cluster node to get data and write data to the cache.

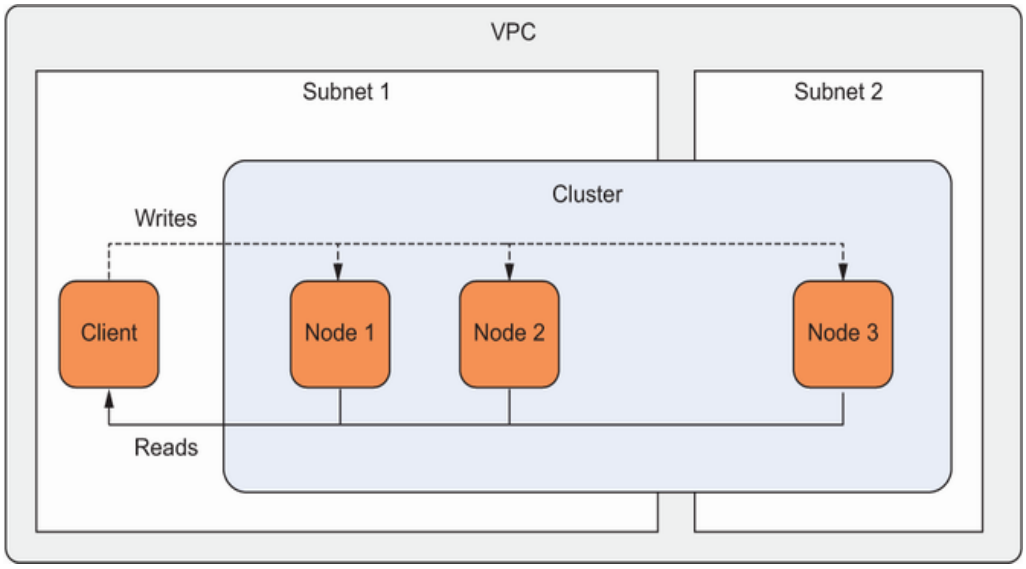


Figure 11.4 Memcached deployment option: Cluster

Use a Memcached cluster if your application requires a simple in-memory store and can tolerate the loss of a node and its data. For instance, the SQL cache example in the beginning of this chapter could be implemented using Memcached. Because the data is always available in the relational database, you can tolerate a node loss, and you need only simple commands (`GET`, `SET`) to implement the query cache.

11.2.2 Redis: Single-node cluster

An ElastiCache for Redis single-node cluster always consists of one node. Sharding and high availability are not possible with a single node. But Redis supports the creation of backups and also allows you to restore those backups. Figure 11.5 shows a Redis single-node cluster.